

Kevin D. Ricks

Curriculum Vitae

Department of Ecology and Evolutionary Biology
University of Toronto

July 2026
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PROFESSIONAL APPOINTMENTS

Postdoctoral Fellow Supervisor: Megan Frederickson	University of Toronto	2023–Current
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EDUCATION

PhD Ecology, Evolution, and Conservation Biology Supervisor: Anthony Yannarell	University of Illinois at Urbana-Champaign	2023
MS Biology Supervisor: Roger Koide	Brigham Young University	2018
BS Environmental Science (<i>cum laude</i>) Minor: Statistics	Brigham Young University	2016

PUBLICATIONS

† = Co-first authors; *Student mentee

In Review:

Ricks KD†, Schwartz C†*, Blaszyński M*, Gonzalez D*, Heath KD, Lau JA & Yannarell AC (In review, PNAS). Mutualisms as engines for rapid adaptation: *Rhizobium* evolution facilitates plant drought resistance. Preprint available at: doi.org/10.64898/2026.07.14.738479

Bedwell SL*, Lakis IM, Megow AR, **Ricks KD**, Lau JA, Fields CJ, Whitaker RJ, & Heath KD (In review, ISME). Ecological shifts within *Rhizobium* communities limit plant health under long-term N-addition. Preprint available at: doi.org/10.64898/2025.12.02.691918

Ricks KD, Bhatt K*, & Frederickson ME (In review, ISME). Stress adaptation of free-living microbes generates novel benefits to plant hosts. Preprint available at: doi.org/10.64898/2026.06.01.729403

Published:

Ricks KD, Raglin SS, & Kent AD (2026). Signature of plant nitrogen adaptation in the associated root-microbiome across genotype panel of *Brachypodium distachyon*. New Phytologist. doi.org/10.1111/nph.70684

Ricks KD, Ricks NJ, & Yannarell AC (2023). Patterns of plant salinity adaptation depend on interactions with soil microbes. The American Naturalist. doi.org/10.1086/725393

Ricks KD & Yannarell AC (2023). Soil moisture incidentally selects for microbes that facilitate locally adaptive plant response. *Proceedings of the Royal Society B*.
doi.org/10.1098/rspb.2023.0469

Nettles R, **Ricks KD**, & Koide RT (2020). The dynamics of antibiotic-induced dysbiosis in interacting bacterial and fungal communities of the mouse colon. *Microbial Ecology*.
doi.org/10.1007/s00248-020-01525-6

Ricks KD & Koide RT (2019). The role of inoculum dispersal and plant species identity in the assembly of leaf endophytic fungal communities. *PLOS ONE*.
doi.org/10.1371/journal.pone.0219832

Ricks KD & Koide RT (2019). Biotic filtering of endophytic fungal communities in *Bromus tectorum*. *Oecologia*. doi.org/10.1007/s00442-019-04388-y

Koide RT, Nguyen BT, Skinner RH, Dell CJ, Adler PR, Drohan PJ, Peoples MS, Licht M, Boyer M, Nettles R, **Ricks KD**, & Watkins J (2018). Comparing biochar application methods for switchgrass yield and C sequestration on contrasting marginal lands in Pennsylvania, USA. *BioEnergy Research*. doi.org/10.1007/s12155-018-9940-1

Koide RT, **Ricks KD**, & Davis ER (2017). Climate and dispersal influence the structure of leaf fungal endophyte communities of *Quercus gambelii* in the eastern Great Basin, USA. *Fungal Ecology*. doi.org/10.1016/j.funeco.2017.08.002

Koide RT, Watkins J, **Ricks KD**, Aranda E, Nettles RM, Yokum H, Yin N, & Clark E (2017). Site and tree factors determining the distribution of *Phellinus tremulae* in *Populus tremuloides* in Utah, USA. *Canadian Journal of Forest Research*. doi.org/10.1139/cjfr-2017-0323

Nettles R, Watkins J, **Ricks KD**, Boyer M, Licht M, Atwood LW, Peoples M, Smith RG, Mortensen DA, & Koide RT (2016). Influence of pesticide seed treatments on rhizosphere fungal and bacterial communities and leaf fungal endophyte communities in maize and soybean. *Applied Soil Ecology*. doi.org/10.1016/j.apsoil.2016.02.008

Szink I, Davis EL, **Ricks KD**, & Koide RT (2016). New evidence for broad trophic status of leaf endophytic fungi of *Quercus gambelii*. *Fungal Ecology*. doi.org/10.1016/j.funeco.2016.04.003

In preparation:

Ricks KD & Frederickson ME. Microbial mutualisms shape modern range shifts. Target journal: *Global Change Biology*.

Ricks KD, Dubey S*, Brissette M*, & Frederickson ME. Incorporating microbial evolution into plant community dynamics. Target journal: *Ecology Letters*.

AWARDS & FELLOWSHIPS

University of Toronto, Arts and Science Postdoctoral Fellowship	CA\$120,000	2023–2025
UIUC PEEC Summer Research Award	\$3,000	2019–2021

Harley J. Van Cleave Research Award	\$1,000	2019
UIUC PEEC Travel Grant	\$1,000	2019–2021
UIUC Graduate College Dissertation Grant	\$3,764	2019–2020
Illinois Distinguished Fellowship	\$75,000	2018–2021
BYU Department of Biology Travel Grant	\$700	2017

TEACHING AND MENTORING

Instructor of record

<i>University of Toronto</i>	2026
BIO 251: Form, Function and Development in Plants	(upcoming)

Teaching Assistant

<i>University of Illinois</i>	2022–2023
NRES 475: Environmental Microbiology	
• Developed and delivered lectures for the course module on Microbial Symbiosis	

<i>Brigham Young University</i>	2014–2015, 2017
BIO 100: Introductory Biology	

Undergraduate mentees

University of Toronto

Kasturi Bhatt	Makenna Brissette
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University of Illinois

David Graves	Alexis Rhoades	Faith Smith
Chase Schwartz	Meghan Blaszyński	Gary Liu
Destiny Gonzalez	Yunqian Zhu	Jake Jagiello

Brigham Young University

Eliza Clark	Emily Aranda	Kate Harmer
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INVITED PRESENTATIONS

- 2026 **Biological Integration Institute: Genomics of eco-evolutionary multiscale symbioses.** Scaling microbial evolution toward plant community dynamics. Champaign, IL, USA.
- 2026 **Canadian Society for Ecology and Evolution.** Incorporating microbial evolution into plant community dynamics. Toronto, ON, Canada.
- 2024 **Canadian Society for Ecology and Evolution.** Symbiosis as engines for rapid adaptation: rhizobia evolution facilitates plant adaptation. Vancouver, BC, Canada.

- 2024 **Duckweeds, microbes, & symbiosis working group.** Using single strains to model plant-microbe interactions. Vancouver, BC, Canada.
- 2023 **University of Toronto.** Rooting microbes in plant adaptation: the evolution of locally adaptive plant-microbe interactions. Toronto, ON, Canada.
- 2020 **Biological Integration Institute seminar series: Genomics of eco-evolutionary multiscale symbioses.** Microbes mediating adaptive plant phenotypes. Virtual.

CONTRIBUTED PRESENTATIONS

- 2026 **Ricks KD & Frederickson ME.** Microbial mutualisms shape modern range shifts. *Evolution*. Cleveland, OH, USA.
- 2025 **Ricks KD & Frederickson ME.** Plant-microbe symbiosis mediates local adaptation in duckweed. *Evolution*. Athens, GA, USA.
- 2023 **Ricks KD, Heath KD, & Yannarell AC.** Symbioses as engines for adaptation? Rhizobia evolution facilitates plant adaptation. *Symposium for Genomics of eco-evolutionary multiscale symbioses*. Bloomington, IN, USA.
- 2022 **Ricks KD, Heath KD, & Yannarell AC.** Evolution of rhizobia populations alter plant adaptation to the moisture environment. *Ecological Society of America*. Montreal, QC, Canada.
- 2021 **Ricks KD & Yannarell AC.** Plant adaptation to local environment through microbial interactions. *Ecological Society of America*. Virtual.
- 2020 **Ricks KD & Yannarell AC.** Adaptive changes with rhizosphere microbial communities to historical watering conditions. *Ecological Society of America*. Virtual.
- 2019 **Ricks KD & Koide RT.** Biotic filtering of endophytic fungal communities in *Bromus tectorum*. *Ecological Society of America*. Louisville, KY, USA.
- 2017 **Ricks KD, Davis ER, & Koide RT.** Climate and dispersal influence the structure of leaf endophyte communities of *Quercus gambelii* in the eastern Great Basin. *Ecological Society of America*. Portland, OR, USA.

Contributions by mentees

- 2026 Brissette M, **Ricks KD & Frederickson ME.** Investigating host, environmental, and microbial modulation of priority effects in the *Lemna* microbiome. *Canadian Society for Ecology and Evolution*. Toronto, Canada.
- 2024 Bhatt K, **Ricks KD, & Frederickson ME.** Experimental evolution of nitrogen fixing bacteria under salt stress. *Center for Global Change*. Toronto, Canada.

- 2024 Schwarz C, **Ricks KD**, Vereau Gorbitz D, Yannarell AC, & Heath KD.. Standing variation or de novo mutation: What drives drought tolerance in experimentally evolved *Rhizobium leguminosarum*? *Lake Arrowhead Microbial Genomics*. Lake Arrowhead, USA.
- 2023 Blaszyński M, Gonzalez D, **Ricks KD**, & Heath KD. Swinging Bacteria: Impacts of *Rhizobium* on nonhost plant species (*Medicago*). *Symposium for Genomics of eco-evolutionary multiscale symbioses*. Urbana, USA.
- 2023 Gonzalez D, Blaszyński M, **Ricks KD**, & Heath KD. Drought mitigating rhizobia. *Illinois Undergraduate Research Symposium*. Urbana, USA.
- 2023 Rhoades A, **Ricks KD**, Wilson P, & Cáceres CE. The impact of diet and disease on microbiome composition in *Daphnia dentifera*. *Illinois Undergraduate Research Symposium*. Urbana, USA.
- 2022 Bedwell SL, **Ricks KD**, Whitaker RJ, & Heath KD (2022). Exploring plasmid stability in a nested legume-rhizobia mutualism undergoing nitrogen addition. *North American Symbiotic Nitrogen Fixation Conference*. Madison, USA.
- 2021 Smith F, **Ricks KD**, & Yannarell AC. Biofilm production in rhizobia influences clover drought response. *Illinois Undergraduate Research Symposium*. Urbana, USA.

SERVICE & OTHER CONTRIBUTIONS

Department Service

UIUC Ecology and Evolution Graduate Admissions Committee	2021–2022
UIUC Ecology and Evolution Steering Committee	2021–2022

Reviewer

Science, National Science Foundation, Ecology, Evolution Letters, ISME, Proceedings of the Royal Society B, Annals of Botany, Functional Ecology, Microbial Ecology, Plant and Soil, Scientific Reports, Frontiers in Plant Science, Ecology and Evolution, Theoretical Ecology, Ecology, New Phytologist

Media Feature

- **National Science Foundation**-Soil microbes help plants cope with drought, but not how scientists thought

Other

- Mentor for PRECS (Phenotypic Plasticity Research Experience for Community College Students) program at UIUC 2021
<https://precs.igb.illinois.edu/>
Facilitated research experiences for community college students
- Mentor for Merit program at UIUC 2022–2023
<https://merit.illinois.edu/mpn/>

Provided undergraduate students from underrepresented backgrounds
mentorship in career and research development

- Participant in the 500 Queer Scientists Visibility Campaign